ACCRA-FLOW

A Modular Geospatial AI Framework for Paratransit Optimization and Urban Resilience in Greater Accra

Adaboh Kwame Twum · **Cajetan Songwae** · **Shaun Esua-Mensah** · **Pout Chop Madeng**

Microsoft Mentors: Moses Igbinigie & Elan Jennings-Jones | Faculty Advisor: Dr. Nana Kwasi Karikari | April 2026

Microsoft × Ashesi Future Builders AI Skilling Program | Case Study Group 5

THE PROBLEM

Every day, Accra grinds to a halt.

Not because the city lacks roads — because it lacks information.

$301.5M
lost annually to congestion

8.21%
of Ghana's GDP

10,000
road deaths per year

Drivers

Choose routes by instinct — zero real-time demand data

Signals

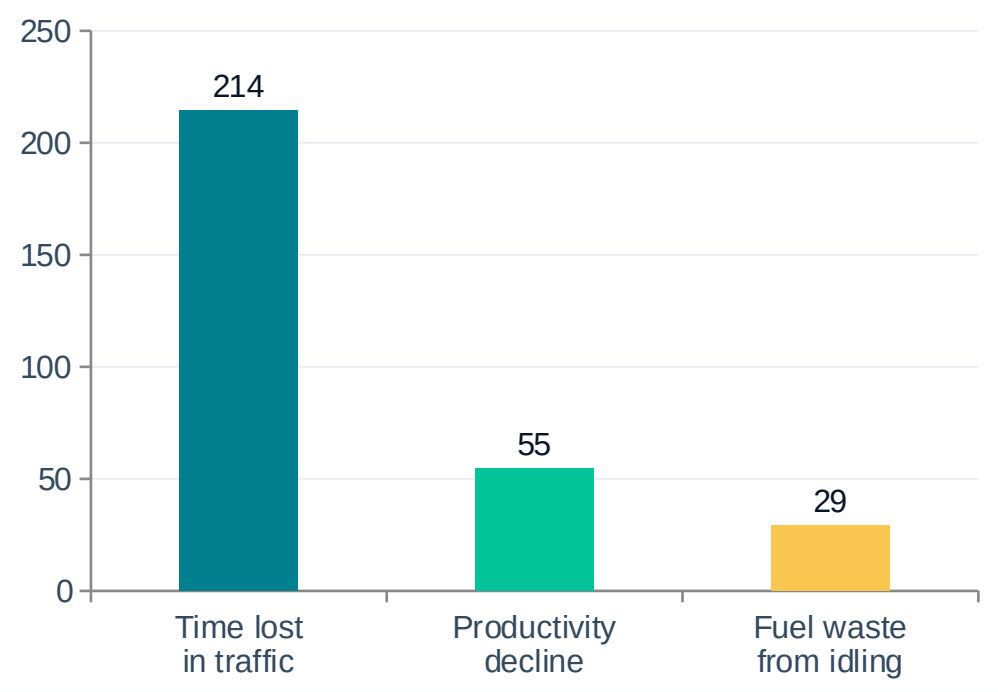
Treat a 28-seat TroTro the same as a private saloon

Commuters

No arrival estimates. No reliable journey planning.

ECONOMIC EMERGENCY

$301.5M Lost Annually — Where Does It Go?



71% — Time Lost

$214.3M / year stolen from workers, students, and families in wasted hours.

18% — Productivity

$54.6M in stress-driven output loss.

11% — Fuel Waste

$29.1M burned while vehicles idle.

Plus: $1.6B in health costs from air pollution. PM2.5 levels in Kaneshie reach 10× WHO limits.

HUMAN IMPACT

Four Lives. One Broken System.

Abena

Student, Univ. of Ghana

10 hrs/week

Lost to unpredictable journey times

Auntie Efua

Trader, Makola Market

14.4%

Post-harvest loss from transport delays

Kojo

TroTro Driver

~40%

Of daily revenue spent on idling fuel

Mensah

Private Commuter

15%

Productive time lost to avoidable congestion

These archetypes became Accra-Flow's design constraints — every technical decision was tested against their reality.

THE SOLUTION

Accra-Flow: Shared Intelligence for a City That Already Works

Three actors. One information layer.



Drivers

Real-time demand signals via USSD — works on any handset



Signals

AI intersections that prioritise a loaded TroTro over an empty car



Commuters

Reliable journey estimates. Finally.

The infrastructure was always there. What was missing was the intelligence to coordinate it.

TECHNOLOGY STACK

Three Tiers. Privacy-First. Built for Accra.

01 PERCEPTION

- NVIDIA Jetson edge units at intersections
- On-device video: counts, type, velocity
- Zero raw footage transmitted — privacy by design
- Output: anonymised metadata only

02 REASONING

- MARL agents — each intersection learns & cooperates
- Pressure propagation solves the Balloon Effect
- 5–30 min ahead traffic forecasting
- SUMO + NVIDIA Omniverse digital twin training

03 INTERACTION

- USSD alerts — no smartphone, no data plan needed
- MoMo Fuel Rebate: AI takes the financial risk
- OpenStreetMap for informal TroTro stop data
- Offline fallback: safe signal cycle on outage

DESIGN EVOLUTION

From Metric to Intelligence: Three Iterations

ITERATION 1

Commuter Fluidity Index

$$CFI = \Sigma(P_i \times V_i) / (T_wait \times C_fuel)$$

WIN: TroTro = 15× value of a private car

LIMIT: Balloon Effect: fixed one spot, broke another

ITERATION 2

TroTro Priority Algorithm

Dynamic Priority Score (occupancy + starvation + touting)

WIN: Smart signals prioritise people, not vehicles

LIMIT: Static rules can't adapt beyond design envelope

ITERATION 3

Multi-Agent Reinforcement Learning

$$R = w_1(CFI) + w_2(-D_queue) + w_3(-E_emissions) - \varphi(v_drift)$$

WIN: Network learns. Balloon Effect solved by design.

LIMIT: Current focus: pilot validation on N4 corridor

PILOT METRICS

Success Targets — Madina–Tetteh Quarshie N4 Corridor

67→48-54	≥60%	40%→34-36%	15-22%
Peak journey time (≥20% reduction)	Driver enrolment by Month 3	Fuel cost as % of revenue	Throughput improvement

Evaluation Timeframe

Month 0–1	Month 2–3	Month 4–6	Month 7+
Baseline collection & USSD load testing	Jetson units live First USSD recommendations	Full metric evaluation Ghost Bus audit	Phase 3/4 expansion (if ≥4/8 metrics met)

RESPONSIBLE AI

Ethics Engineered In — Not Added On

Privacy-by-Architecture

All video processed on-device. Zero raw footage transmitted. Compliant with Ghana Data Protection Act 2012.

Anti-Bias Design

Person-weighted CFI prevents over-prioritising private cars. Starvation penalty stops side-streets from being ignored.

Digital Inclusion

USSD = no smartphone required. MoMo penetration far exceeds smartphone ownership. No one is excluded.

Safety Guardrails

Minimum all-red clearance enforced. 90-sec communication loss → reverts to safe TPA baseline automatically.

Explainability

Plain-language USSD messages include reasoning. E.g. "High demand at Madina — fuel credit GH₵2.00 applied."

Human Override

Transport authority operators retain full manual override. AI assists; humans decide.

ECOSYSTEM

Partners, Budget & Path to Deployment

Ghana Highways Authority / GAMA — ESSENTIAL

Road access rights & signal infrastructure

MTN Ghana / Telecel Ghana — ESSENTIAL

USSD gateway + MoMo API integration

TroTro Driver Unions (GPRTU) — ESSENTIAL

Driver recruitment, trust & co-design

NVIDIA (Inception Program) — HIGH

Jetson hardware at research rates

Microsoft (AI for Good) — HIGH

Azure compute credits for MARL training

Ashesi University — HIGH

Research validation & field data team

Phase 1–2 Capital Requirement: $8,000–$12,000 USD | Largely coverable via in-kind + NVIDIA Inception + Microsoft AI for Good + MTN CSI

PROJECTED IMPACT

What Accra Gets Back

$45–72M

Recoverable annual economic loss (Phase 1–2)

20–25%

Reduction in peak journey times

15–22%

Air quality improvement (PM2.5)

GH₵ back

To drivers via MoMo fuel rebate tokens

The $301.5 million to be reclaimed is not merely a financial figure.
It represents 10 hours/week returned to students like Abena. Perishable inventory preserved for Auntie Efua. Fuel margins restored to drivers like Kojo.

When urban mobility fails, it fails people. When it succeeds, it returns the most finite resource they have: **time.**

WHAT'S NEXT

Three Things That Unlock Accra-Flow's Potential

1

Institutional Access

MoU with Ghana Highways Authority for Jetson unit installation on the N4 corridor.

2

Driver Partnership

GPRTU co-design of USSD UX and MoMo rebate — before we ask drivers to change anything.

3

Compute & Hardware

Azure AI for Good credits + NVIDIA Inception hardware subsidy to launch Phase 1 baseline.

Accra-Flow | Case Study Group 5 | Microsoft × Ashesi Future Builders AI Skilling Program | April 2026

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